

Lab 16. Getting started with LocalChat

(Should be Executed on your own Machine)

You get to decide the setup for these 3 main components:

1. **LLM**: the large language model provider used for inference. It can be local, or remote, or even OpenAI.
2. **Embeddings**: the embeddings provider used to encode the input, the documents and the users' queries. Same as the LLM, it can be local, or remote, or even OpenAI.
3. **Vector store**: the store used to index and retrieve the documents.

Sequence 1. Installing Ollama

a) Installing on Linux

- i. Login to your linux machine as root user and launch the terminal.
- ii. Run the following command from terminal

```
curl -fsSL https://ollama.com/install.sh | sh
```

b) Installing on Windows

- i. Download the following installer for windows

```
https://ollama.com/download/OllamaSetup.exe
```

- ii. Run the Installer and follow the instructions

c) Installing on MacBook (M1/M2/M3 Chipset)

- i. Download the Installer

```
https://ollama.com/download/Ollama-darwin.zip
```

- ii. Extract the Installer and Run.

Sequence 2. Installing LocalChat

1. Create a Python Environment with Python version 3.11 with pyenv or conda:

```
# curl https://pyenv.run | bash
```

a. Using pyenv

- Check the default version set by **pyenv**. The *** indicates that the system Python version is active** currently.

```
$ pyenv versions
```

```
* system (set by /home/opc/.pyenv/version)
3.11.9
```

- If you see **"system"**, this means that, by default, you are still using your system Python:

```
$ python -V
```

- Change the default global version to python 3.11.9 with:

```
$ pyenv global 3.11.9
```

```
$ pyenv versions
```

```
system
* 3.11.9 (set by /home/opc/.pyenv/version)
```

b. With conda

- Check default Python Version (No Active Environment):

```
$ Python -V
```

```
Python 3.12.6
```

- Activate base environment and check Python Version:

```
info@sangwan ~/labs$ conda activate base
```

```
(base) info@sangwan ~/labs$ Python -V
```

```
Python 3.12.8
```

- Create an environment with Python 3.11 version:

```
$ conda create -n LocalChat Python=3.11.9
```

- Activate New environment and check Python Version:

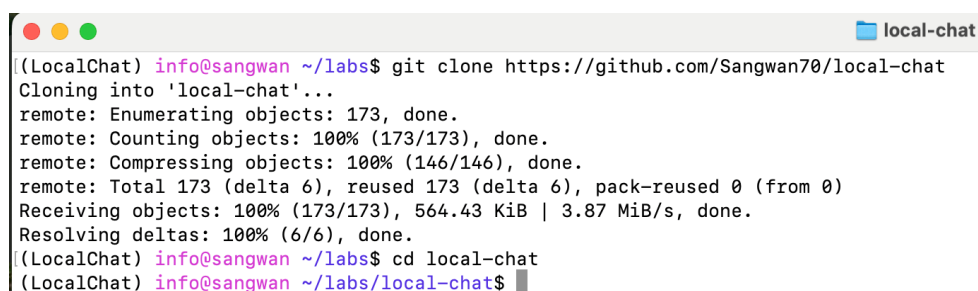
```
$ conda activate LocalChat
```

```
(LocalChat) info@sangwan ~/labs$ python -V
```

```
Python 3.11.9
```

2. Clone the LocalChat Repository and change to the downloaded directory:

```
git clone https://github.com/Sangwan70/local-chat
cd local-chat
```



```
local-chat
(LocalChat) info@sangwan ~/labs$ git clone https://github.com/Sangwan70/local-chat
Cloning into 'local-chat'...
remote: Enumerating objects: 173, done.
remote: Counting objects: 100% (173/173), done.
remote: Compressing objects: 100% (146/146), done.
remote: Total 173 (delta 6), reused 173 (delta 6), pack-reused 0 (from 0)
Receiving objects: 100% (173/173), 564.43 KiB | 3.87 MiB/s, done.
Resolving deltas: 100% (6/6), done.
(LocalChat) info@sangwan ~/labs$ cd local-chat
(LocalChat) info@sangwan ~/labs/local-chat$
```

3. Install Poetry for dependency management.

```
pip install -qU poetry
```

4. Setup for Windows (**Skip to step 5 for Linux/Mac**)

- a. Enable Hyper-V (if not enabled already) from **Control Panel - > Programs and Applications -> Enable/Disable Windows Features** and Restart your computer.
- b. Install WSL ((Windows Subsystem for Linux) to access the power of both Windows and Linux at the same time on a Windows machine. Visit the following page and follow instructions:

<https://learn.microsoft.com/en-us/windows/wsl/install>

- c. Launch Powershell as administrator and execute following commands to install dependencies:

```
Set-ExecutionPolicy Bypass -Scope Process -Force;  
[System.Net.ServicePointManager]::SecurityProtocol =  
[System.Net.ServicePointManager]::SecurityProtocol -bor 3072; iex ((New-Object  
System.Net.WebClient).DownloadString('https://community.chocolatey.org/install.ps1'))
```

(Visit https://www.reddit.com/r/chocolatey/comments/1928qc9/anyone_else_having_issues_installing_chocolatey/ for the exact command)

```
choco install make -y  
choco upgrade chocolatey
```

- d. Close the current window and Open the Project directory from terminal (cmd). Install Extra Packages required to run LocalChat

```
poetry add fastapi uvicorn  
poetry install --extras embeddings-ollama  
poetry install --extras vector-stores-qdrant  
poetry install --extras llms-llama-cpp  
poetry install --extras llms-ollama  
poetry install --extras ui
```

- e. Run the LocalChat

```
set PGPT_PROFILES=ollama  
make run
```

5. Open **another Tab** of your terminal and **Start Ollama** to start a local inference server, serving both the LLM and the Embeddings:

```
ollama serve
```

If you get the following error, that would mean ollama is already running.

Error: listen tcp 127.0.0.1:11434: bind: address already in use

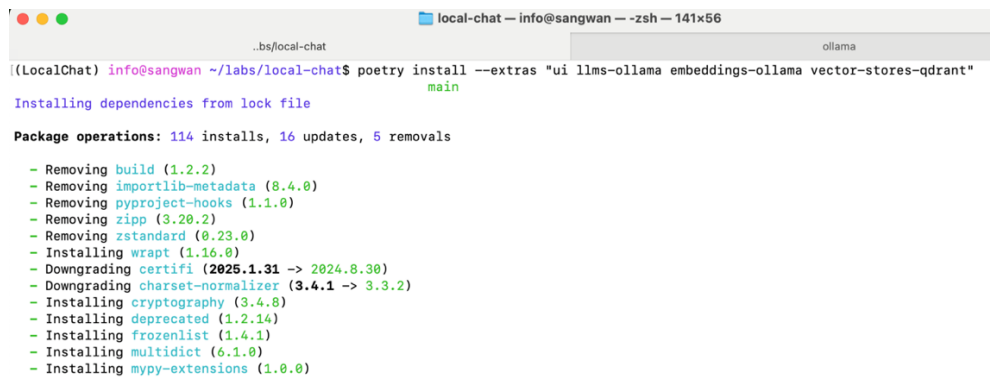
```

info@MacBook-Pro private-gpt % pip install -qU poetry
info@MacBook-Pro private-gpt % ollama serve
2024/12/05 22:14:54 routes.go:1197: INFO server config env="map[HTTPS_PROXY: HTTP_
PU_OVERHEAD:0 OLLAMA_HOST:http://127.0.0.1:11434 OLLAMA_KEEP_ALIVE:5m0s OLLAMA_LLM
QUEUE:512 OLLAMA_MODELS:/Users/info/.ollama/models OLLAMA_MULTIUSER_CACHE:false OL
IGINS:[http://localhost https://localhost http://localhost:* https://localhost:* h
http://0.0.0.0 https://0.0.0.0 http://0.0.0.0:* https://0.0.0.0:* app://* file://*
ttp_proxy: https_proxy: no_proxy:]"
time=2024-12-05T22:14:54.539+05:30 level=INFO source=images.go:753 msg="total blob
time=2024-12-05T22:14:54.540+05:30 level=INFO source=images.go:760 msg="total unus
time=2024-12-05T22:14:54.542+05:30 level=INFO source=routes.go:1248 msg="Listening
time=2024-12-05T22:14:54.542+05:30 level=INFO source=common.go:135 msg="extracting
ama3734876440/runners
time=2024-12-05T22:14:54.595+05:30 level=INFO source=common.go:49 msg="Dynamic LLM
time=2024-12-05T22:14:54.616+05:30 level=INFO source=types.go:123 msg="inference c
="21.3 GiB" available="21.3 GiB"

```

6. The default `settings-ollama.yaml` is configured to use llama3.2 LLM (~2GB) and nomic-embed-text Embeddings (~275MB)
7. Back to previous Tab of your terminal, install LocalChat with:

```
poetry install --extras "ui llms-ollama embeddings-ollama vector-stores-qdrant"
```



```

..local-chat
local-chat - info@sangwan -- zsh - 141x56
ollama
(LocalChat) info@sangwan ~/labs/local-chat$ poetry install --extras "ui llms-ollama embeddings-ollama vector-stores-qdrant"
main
Installing dependencies from lock file

Package operations: 114 installs, 16 updates, 5 removals

- Removing build (1.2.2)
- Removing importlib-metadata (8.4.0)
- Removing pyproject-hooks (1.1.0)
- Removing zipp (3.20.2)
- Removing zstandard (0.23.0)
- Installing wrapt (1.16.0)
- Downgrading certifi (2025.1.31 -> 2024.8.30)
- Downgrading charset-normalizer (3.4.1 -> 3.3.2)
- Installing cryptography (3.4.8)
- Installing deprecated (1.2.14)
- Installing frozenlist (1.4.1)
- Installing multidict (6.1.0)
- Installing mpy-extensions (1.0.0)

```

8. Now run LocalChat. (Make sure you have a working Ollama running locally.)

```
PGPT_PROFILES=ollama make run
```

```

22:19:05.400 [INFO] ] private_gpt.components.llm.llm_component - Initializing the LLM in mode=ollama
22:19:05.486 [INFO] ] httpx - HTTP Request: GET http://localhost:11434/api/tags "HTTP/1.1 200 OK"
22:19:05.488 [INFO] ] httpx - HTTP Request: GET http://localhost:11434/api/tags "HTTP/1.1 200 OK"
22:19:06.636 [INFO] ] private_gpt.components.embedding.embedding_component - Initializing the embedding model in mode=ollam
22:19:06.677 [INFO] ] httpx - HTTP Request: GET http://localhost:11434/api/tags "HTTP/1.1 200 OK"
22:19:06.679 [INFO] ] httpx - HTTP Request: GET http://localhost:11434/api/tags "HTTP/1.1 200 OK"
22:19:06.679 [INFO] ] llama_index.core.indices.loading - Loading all indices.
22:19:06.679 [INFO] ] private_gpt.components.ingest.ingest_component - Creating a new vector store index
Parsing nodes: 0it [00:00, ?it/s]
Generating embeddings: 0it [00:00, ?it/s]
22:19:16.377 [INFO] ] private_gpt.ui.ui - Mounting the gradio UI, at path=/
22:19:16.828 [INFO] ] uvicorn.error - Started server process [6180]
22:19:16.828 [INFO] ] uvicorn.error - Waiting for application startup.
22:19:16.828 [INFO] ] uvicorn.error - Application startup complete.
22:19:16.829 [INFO] ] uvicorn.error - Uvicorn running on http://0.0.0.0:8001 (Press CTRL+C to quit)

```

If you get any error related to Module Not found. . Install the module with pip. For example "pip install -qU build" and run the above command again.

```

(LocalChat) info@sangwan ~/labs/local-chat$ PGPT_PROFILES=ollama make run
poetry run python -m local_chat

No module named 'build'
make: *** [run] Error 1

```

```

(LocalChat) info@sangwan ~/labs/local-chat$ pip install -qU build
(LocalChat) info@sangwan ~/labs/local-chat$ PGPT_PROFILES=ollama make run
poetry run python -m local_chat
18:28:06.666 [INFO] ] local_chat.settings.settings_loader - Starting application with profiles=['default', 'ollama']
None of PyTorch, TensorFlow >= 2.0, or Flax have been found. Models won't be available and only tokenizers, configuration and f
ties can be used.
18:28:23.611 [INFO] ] local_chat.components.llm.llm_component - Initializing the LLM in mode=ollama
18:28:23.653 [INFO] ] httpx - HTTP Request: GET http://localhost:11434/api/tags "HTTP/1.1 200 OK"
18:28:23.655 [INFO] ] httpx - HTTP Request: GET http://localhost:11434/api/tags "HTTP/1.1 200 OK"
18:28:26.243 [INFO] ] local_chat.components.embedding.embedding_component - Initializing the embedding model in mode=ollama
18:28:26.262 [INFO] ] httpx - HTTP Request: GET http://localhost:11434/api/tags "HTTP/1.1 200 OK"
18:28:26.265 [INFO] ] httpx - HTTP Request: GET http://localhost:11434/api/tags "HTTP/1.1 200 OK"
18:28:26.265 [INFO] ] llama_index.core.indices.loading - Loading all indices.
18:28:26.265 [INFO] ] local_chat.components.ingest.ingest_component - Creating a new vector store index
Parsing nodes: 0it [00:00, ?it/s]
Generating embeddings: 0it [00:00, ?it/s]
18:28:53.537 [INFO] ] local_chat.ui.ui - Mounting the gradio UI, at path=/
18:28:54.011 [INFO] ] uvicorn.error - Started server process [3768]
18:28:54.011 [INFO] ] uvicorn.error - Waiting for application startup.
18:28:54.012 [INFO] ] uvicorn.error - Application startup complete.
18:28:54.012 [INFO] ] uvicorn.error - Uvicorn running on http://0.0.0.0:8001 (Press CTRL+C to quit)

```

- LocalChat will use `settings-ollama.yaml` settings file, which is already configured to use Ollama LLM and Embeddings, and Qdrant.

Launch the browser and launch the UI (available at <http://localhost:8001>)

MY LOCAL CHATBOT

Mode: ☒ RAG ☐ Local Search ☐ Basic

☐ Summarize

Get contextualized answers from selected files.

Upload File(s)

Ingested Files

File name

De-select selected file

Selected for Query or Deletion

All files

Delete selected file

Delete ALL files

LLM: ollama | Model: llama3.2

Retry Undo Clear

Type a message...

Submit

Additional Inputs

Mode

☐ RAG
 ☐ Local Search
 ☒ Basic

☐ Summarize

Chat with the LLM using its training data. Files are ignored.

Upload File(s)

Ingested Files

File name

responsible-ai-oracle.pdf

De-select selected file

Selected for Query or Deletion

All files

Delete selected file

Delete ALL files

10. Select RAG and set the System Prompt to:

"You are a helpful AI assistant. Use the following pieces of context to answer the question at the end. If you don't know the answer, just say you don't know. DO NOT try to make up an answer. If the question is not related to the context, politely respond that you are tuned to only answer questions that are related to the context."

Additional Inputs

System Prompt

You are a helpful AI assistant. Use the following pieces of context to answer the question at the end. If you don't know the answer, just say you don't know. DO NOT try to make up an answer.

If the question is not related to the context, politely respond that you are tuned to only answer questions that are related to the context.

11. Ask a Question "How Will Responsible AI Impact the Future of Work?"

MY LOCAL CHATBOT

Mode

☒ RAG
 ☐ Local Search
 ☐ Basic

☐ Summarize

Get contextualized answers from selected files.

Upload File(s)

Ingested Files

File name

responsible-ai-oracle.pdf

De-select selected file

Selected for Query or Deletion

All files

Delete selected file

Delete ALL files

LLM: ollama | Model: llama3.2

How Will Responsible AI Impact the Future of Work?

According to the provided text, responsible AI has the potential to impact the future of work in several ways:

1. Reduced bias and discrimination: By using fair algorithms and data, responsible AI can help reduce bias and discrimination in hiring and other workplace decisions.
2. Increased efficiency and productivity: AI can automate many routine tasks and decision-making processes, allowing workers to focus on more complex and creative work.

These effects have the potential to create a more diverse and inclusive workforce, which has been shown to improve business outcomes.

Sources:

1. responsible-ai-oracle.pdf (page 9)

Retry

Undo

Clear

Type a message...

Submit

Additional Inputs

System Prompt

You are a helpful AI assistant. Use the following pieces of context to answer the question at the end. If you don't know the answer, just say you don't know. DO NOT try to make up an answer.

If the question is not related to the context, politely respond that you are tuned to only answer questions that are related to the context.

12. Try asking a question not in the context. For example ask about yourself (I asked Tell me something about Ram N Sangwan)

MY LOCAL CHATBOT

Mode

• RAG

Local Search

Basic

Summarize

Get contextualized answers from selected files.

Upload File(s)

Ingested Files

File name

responsible-ai-oracle.pdf

De-select selected file

Selected for Query or Deletion

All files

Delete selected file

Delete ALL files

LLM: ollama | Model: llama3.2

Tell me something about Ram N Sangwan

I don't know anything about Ram N Sangwan from the provided context information. The text only mentions Dr. Sanjay Basu as the author of "Oracle's Guide to Ethical Considerations in AI Development and Deployment". There is no mention of a person named Ram N Sangwan. If you could provide more context or information, I'd be happy to try and help further!

Sources:

1. responsible-ai-oracle.pdf (page 2)

2. responsible-ai-oracle.pdf (page 17)

Retry

Undo

Clear

Type a message...

Submit

Additional Inputs

System Prompt

You are a helpful AI assistant. Use the following pieces of context to answer the question at the end. If you don't know the answer, just say you don't know. DO NOT try to make up an answer.
If the question is not related to the context, politely respond that you are tuned to only answer questions that are related to the context.